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REVIEW

Is thermography an effective screening tool for differentiating benign and malignant skin lesions in the head and neck? A systematic review

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Abstract
The aim of this study was to assess, through a systematic review, the status of infrared thermography (IRT) as a diagnostic tool for skin neoplasms of the head and neck region and in order to validate its effectiveness in differentiating benign and malignant lesions. A search was carried out in the LILACS, PubMed/MEDLINE, SCOPUS, Web of Science and EMBASE databases including studies published between 2004 and 2024, written in the Latin-Roman alphabet. Accuracy studies with patients aged 18 years or over presenting benign and malignant lesions in the head and neck region that evaluated the performance of IRT in differentiating these lesions were included. Lesions of mesenchymal origin and studies that did not mention histopathological diagnosis were excluded. The systematic review protocol was registered in the PROSPERO database (CRD42023416079). Reviewers independently analyzed titles, abstracts, and full-texts. After extracting data, the risk of bias of the selected studies was assessed using the QUADAS − 2 tool. Results were narratively synthesized and the certainty of evidence was measured using the GRADE approach. The search resulted in 1,587 records and three studies were included. Only one of the assessed studies used static IRT, while the other two studies used cold thermal stress. All studies had an uncertain risk of bias. In general, studies have shown wide variation in the accuracy of IRT for differentiating between malignant and benign lesions, with a low level of certainty in the evidence for both specificity and sensitivity.

Keywords Thermography · Skin temperature · Skin neoplasms · Early detection of cancer

Introduction

The skin is the largest organ in the human body and is considered the first line of defense in protecting the body against ultraviolet radiation [1]. This organ can undergo several changes in its structural components, leading to the development of benign and malignant tumors [2]. Most skin tumors are benign and morphologically manifest as papules, nodules, slow-growing keratotic or cystic lesions. On the other hand, malignant tumors are usually solitary, irregular, fast-growing plaques or nodules that can ulcerate [3].

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The most common malignant tumors arise from the epidermis, including melanoma and non-melanoma cancers such as basal cell carcinoma, squamous cell carcinoma, and Merkel cell carcinoma [4]. These lesions can affect any part of the body; however, they are more frequent in areas exposed to sunlight, such as hands, face, neck, lips and posterior region of the trunk [5, 6]. In 2020, non-melanoma skin cancer (excluding basal cell carcinoma) accounted for more than one million new cases and 64,000 deaths worldwide, with incidence rates approximately 2 times higher among men than among women, while melanoma was responsible for more than 324,000 new cases and 57,043 deaths [7].

Early detection of skin cancers enhances the effectiveness of treatment protocols and improves prognosis [8]. The diagnosis of these lesions is still challenging because many malignant lesions, such as melanoma, can resemble a seborrheic keratosis or a nevus [2]. Visual inspection, followed later by histopathological confirmation, remains the dermatologist's reference test assessment tool for most dermatological conditions; however, visual inspection is a subjective assessment and may be associated with significant interobserver variations [9]. To overcome this limitation, several tools and techniques have been developed to improve the evaluation of skin tumors [4, 9], such as digital photography, dermoscopy, confocal reflectance microscopy, optical coherence tomography, high-frequency ultrasound (US) [4], ultraviolet light photography [9] and infrared thermography (IRT) [10].

IRT is a non-invasive diagnostic method that uses a thermographic camera to record the amount of heat emitted by the skin, not causing any risk to the patient's health, due to the absence of ionizing radiation [10]. This technique can be employed in two ways: without prior stimulus, known as *passive* IRT, or dynamically, by eliciting a response through the application of a chemical, mechanical, or thermal load, referred to as *dynamic* IRT [10]. It is important to note that tumors induce the growth of new capillary blood vessels through the production of specific growth factors that promote angiogenesis. Therefore, the presence of new blood vessels and increased blood supply somehow alters the thermal response of tumor cells when a stimulus is applied, which may leverage the use of IRT for their diagnostic [11].

Several studies have investigated the use of IRT in epithelial neoplasms as a complementary tool in the diagnostic process of skin tumors in various regions of the body and to improve accuracy and precision in decision-making regarding the most appropriate treatment [10–12]. However, to date, no study has systematically assessed the effectiveness of such method as a diagnostic tool for skin neoplasms in the head and neck region. Thus, this systematic review aimed to describe the status of IRT applications in the assessment of skin neoplasms in the head and neck region and to evaluate its effectiveness in differentiating benign and malignant lesions.

Materials and methods

This systematic review is reported according to the protocol recommended by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [13]. The systematic review protocol was registered in the PROSPERO database under the number CRD42023416079.

Eligibility criteria

This systematic review study aimed to answer the following question: "Is IRT accurately capable of differentiating benign and malignant lesions in patients with skin lesions in the head and neck region?". The question was formulated following the PIRD strategy, described below:

P – Patients with skin lesions of epithelial origin in the head and neck.

I – Dynamic or passive infrared thermography.

R – Clinical examination with histopathological confirmation.

D – Differentiation of benign and malignant lesions.

The eligibility criteria are described in Table 1.

Table 1 Inclusion and exclusion criteria defined for inclusion of studies in this systematic review

	Inclusion	Exclusion
P	Adult patients (aged 18 years or older) with lesions located in the skin of the head and neck region (with diagnosis confirmed by histopathological examination).	Patients with lesions in the eye region and elsewhere on the body or with lesions of mesenchymal origin and studies that did not mention the histopathological diagnosis of the lesions.
I	Accuracy studies that evaluated the performance of infrared thermography, dynamic or passive, the interest groups, published from January 2014 to May 2024, written in the Latin-Roman alphabet.	Cross-sectional studies, cohort studies, case reports, case series, literature reviews, letters to the editor / editorials, personal opinions, books / book chapters, textbooks, reports, conference abstracts and patents.
R	Studies that used clinical examination (visual method, palpation).	Studies using other diagnostic methods, such as ultrasonography and confocal microscopy.
D	Differentiation of benign and malignant lesions.	Studies that did not analyze benign and malignant skin lesions in the head and neck region.

Information sources

The Latin-American and Caribbean Health Sciences Literature (LILACS), Medical Publications (PubMed)/ MEDLINE (including electronic publications ahead of print), SCOPUS, Web of Science and EMBASE databases were searched in May 2024.

Search strategy

An electronic search strategy was elaborated using MeSH (Medical Subject Headings) and non-MeSH terms combined with Boolean operators "AND" and "OR". In addition, MeSH entry terms were used. The detailed search strategy used in each database is available in Supplementary Table 1. A search of the grey literature was performed in Google Scholar (the first 100 relevant references). A manual search was also performed through a systematic analysis of the references of eligible articles.

Selection and data collection process

The records obtained from the search were exported to the software Rayyan - AI Powered Tool for Systematic Literature Reviews (Qatar Foundation, Qatar) and duplicates were manually removed.

Studies were selected at three different times. In the first phase, reviewers discussed the eligibility criteria and titles not related to the topic and books/book chapters were eliminated. In the second phase, reviewers read all abstracts independently. At this stage, abstracts not related to the topic were eliminated. In the third search stage, the complete articles were read in full to verify whether they met the eligibility criteria. At all stages, when consensus was not achieved, a third reviewer was consulted to make a final decision. Rejected studies were recorded separately, explaining the reasons for exclusion.

Data items

Two reviewers independently analyzed the studies and extracted the following information: year of publication, country, language, data collection period, study location, age/age group, initial and final sample size, type of lesion, diagnostic test used, type of thermal stress, material used to cause the thermal stress, reference test used, main results and accuracy. Any disagreement between the reviewers was resolved through discussion and when consensus was not reached; a third reviewer was consulted to make the final decision.

In cases where the data was unclear in the study, emails were sent to the authors for clarification. Despite sending emails to the authors of three studies, no responses were received.

Study risk of bias assessment

Two reviewers independently assessed the risk of bias of the selected studies using the QUADAS −2 tool (Quality Assessment of Diagnostic Accuracy Studies) [14]. Risk of bias is classified as high, low or uncertain. Two investigators independently completed the risk of bias assessment and resolved discrepancies to reach a consensus. A third experienced researcher was consulted if discrepancies persisted.

Synthesis methods

Due to clinical and methodological heterogeneity, a narrative synthesis of the results of the sensitivity, specificity and accuracy outcomes was carried out.

Certainty assessment

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) tool [15, 16] was used to assess the certainty of evidence. This assessment was based on study design, inconsistency, indirectness, imprecision, and other considerations. The certainty of evidence can be characterized as high, moderate, low, or very low [15, 16].

Results

Study selection

The search strategies retrieved a total of 1,587 articles. The first 100 results obtained by applying the search keys on Google Scholar were also included. From the assessed studies, 146 duplicates were removed, and 119 studies were marked as ineligible based on a tool in Rayyan software used to identify terms that are not relevant to the search. The titles and abstracts of 1,322 studies were read and 1,290 studies were excluded. Most of the studies were excluded due to their focus on animal tests, literature reviews, mathematical models, or the assessment of eye tissues. Additionally, studies involving systemic chronic diseases such as diabetes and rheumatoid arthritis were considered beyond the scope of this review.

Thirty-two records were considered potentially eligible and their full texts were assessed by the two reviewers (DPM and DFBS). After reading the full texts of the preselected articles, 29 studies did not meet any of the inclusion criteria. Therefore, three papers composed the final sample

[11, 17, 18]. The flowchart depicting the article selection process is shown in Fig. 1.

Description of included studies

The characteristics of the included studies are described in Tables 2 and 3. All the studies were written in English. Two studies were from the United States [11, 18] and one from Brazil [17]. The sample size of the studies varied from 102 to 140 patients, with a higher prevalence of male patients (58%). The number of benign lesions varied from 59 to 155 and the most prevalent lesion was intradermal nevus. On the other hand, the number of malignant lesions varied from 43 to 155 and the most prevalent lesion was basal cell carcinoma. None of the assessed studies mentioned the anatomical location of the lesions when describing the methodology and results.

One of the assessed studies used static IRT [17], while the other two employed dynamic IRT [11, 18]. In studies using dynamic IRT, cold thermal stress was applied using a Ranque-Hilsch vortex tube, followed by a commercially available air conditioning unit [11]. Regarding the reference test, all studies used the visual method with histopathological confirmation. All the included studies used artificial intelligence classifiers to assist in detecting and differentiating lesions.

Synthesis of results

In the two included studies that used dynamic IRT, the cold thermal stress was used, achieved by means of a Ranque-Hilsch vortex tube [11, 18]. Later, in one of the studies, this cooling unit was replaced by a commercial air conditioner unit due to portability and ability to provide a constant flow of conditioned air [11]. However, in the cited study, no cooling temperature was specified for the conditioned air flow [11]. Furthermore, another limitation was that it is difficult to control the outlet temperature of a Ranque-Hilsch vortex tube. Its outlet temperature depends on pressure, flow and ambient temperature, and it is not constant in the initial seconds of the cooling procedure [20]. Another challenge in dynamic IRT is to keep the patient as still as possible during the exam.

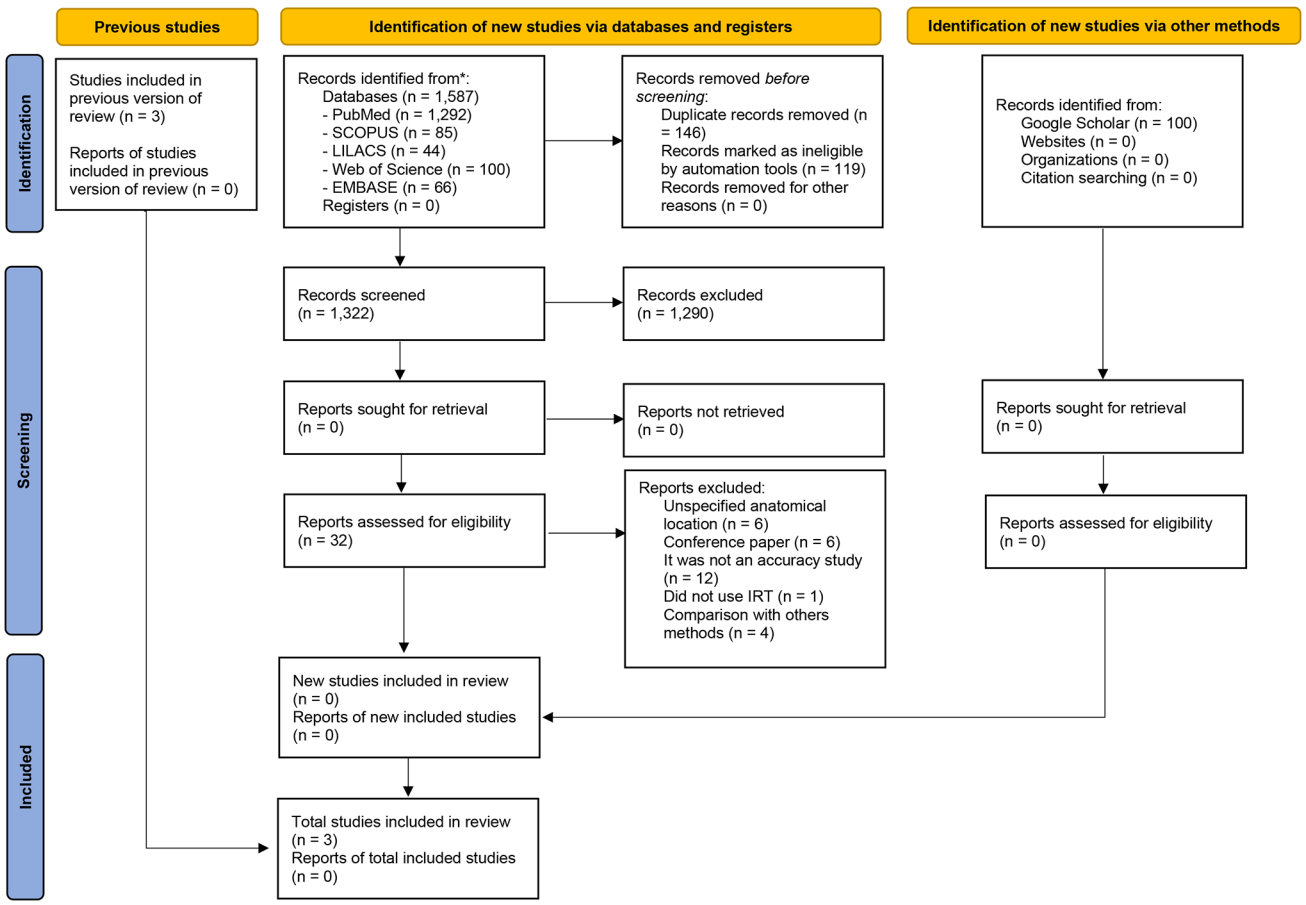


Fig. 1 Screening for registration and selection of included studies. PRISMA three-phase flowchart for study collection showing the number of studies identified, eligible and included in the review (PRISMA 2020)

Table 2 Data extracted from the included articles regarding country, language, sample, sex, type and location of lesions

Authors/ year	Country	Language	Sample size	Sex	Type of lesion	Location of the lesion
Godoy et al.[18]	The United States	English	102	58% male, 42% female	59 benign lesions, 29 basal-cell carcinoma, 8 squamous-cell carcinoma, and 6 malignant melanomas.	-**
Godoy et al.[11]	The United States	English	140	58% male, 42% female	82 benign lesions, 58 malignant lesions (6 malignant melanomas, 42 basal cell carcinomas, 10 with squamous-cell carcinoma).	-**
Stringasci et al.[17]	Brazil	English	-	-	100 basal cell carcinoma, 100 intradermal nevus, 35 squamous cell carcinoma, 35 actinic keratosis, 20 pigmented seborrheic keratosis, and 20 melanomas.	**_

** The studies did not mention the anatomical location of the lesions; however, the available figures identified lesions in the head and neck region

Similar results of sensitivity and specificity were observed in the studies that included dynamic IRT. Godoy et al. [18] reported sensitivity of 95%, specificity of 83%, and area under the ROC curve of 95%. Similarly, the study by Godoy et al. [11] obtained 100% sensitivity and 90% specificity for 93% of 200 training datasets. On the other hand, in the study by Stringasci et al. [17], which evaluated pairs of lesions with similar characteristics, squamous cell carcinoma versus actinic keratosis was the pair of lesions that had the best differentiation (82.9% sensitivity and 85.7% specificity), while melanoma versus pigmented seborrheic keratosis was the pair of lesions with less accuracy (65% sensitivity and 70% specificity).

Two of the included studies in this systematic review explored the application of artificial intelligence classifiers to facilitate the understanding of thermal image visualization and optimize the final diagnosis [11, 17]. Godoy et al. [18] assessed benign (59 samples) and malignant (43 samples) lesions and differentiated them based on the difference in temperature between these two types of lesions. The authors found that the temperature limit that achieved maximum lesion differentiation (sensitivity = 95%; specificity = 83%) was 0.238 °C. In another study by the same group [11], Karhunen-Loève coefficients were recovered from the thermal recovery curves and applied to a statistical algorithm. In Stringasci et al. study [17], an SVM model was used to report sensitivity and specificity of 82.9% and 85.7% for differentiation of squamous cell carcinoma and actinic keratosis, respectively.

Risk of bias appraisal

The results of the risk of bias assessment by QUADAS-2 are presented in Table 4. Regarding patient selection, all the three studies did not show inappropriate exclusions. In addition, patients with malignant and benign skin lesions were included. All the studies mentioned that the reference test for case selection was the visual/photographic method with confirmation of histopathological analysis. In all the studies more than one domain was classified as "unclear" (Supplementary Table 2); therefore, they were considered as at risk of bias with concerns regarding applicability.

Assessment of certainty of the evidence

The certainty of evidence for all the outcomes assessed was very low (Supplementary Table 3). For the sensitivity outcome, the certainty of the evidence was low due to problems in the risk of bias domain, specifically inconsistencies in patient selection, the index test, flow, and timing. Furthermore, the certainty of the evidence was lowered in one level due to problems in inconsistency and imprecision.

Table 3 Type of IT, thermal stress, material, camera and classification algorithms used in the studies, in addition to the reference test, main results and features analyzed in the images

Authors/year	Diagnostic test used	Resolution of the camera used	Type of thermal stress	Material used	Reference test used	Classification algorithms used	Features analyzed in the images	Main results	Accuracy
Godoy et al.[18]	Dynamic IRT	300 microns per pixel	Cold	Rank-Hilsch vortex for 30 s	Visual + histopathological	-	The lesion selection by displaying u(i, j,1) (i.e., the spatially aligned visible image) to the user in order to ask him/her to draw the boundary of the suspicious lesion using Matlab's imellipse function; the malignant tissue when compared with benign tissue should present different thermal parameters such as thermal diffusion, tissue specific heat, metabolic heat generation, etc.	The dynamic IRT approach in conjunction with careful pixel selection has the potential to provide a fast, accurate, contactless and non-invasive way to screen for common types of skin cancer. As such, it has the potential to significantly reduce the number of biopsies performed on similar lesions.	Sensitivity and specificity were 95% (with a 95% confidence interval of [87.8% 100.0%]) and 83% (with a 95% confidence interval of [73.4% 92. 5%]), respectively, and with an area under the ROC curve of 95%.
Godoy et al.[11]	Dynamic IRT	300 microns per pixel	Cold	Ranque-Hilsch vortex tube, followed by a commercially available air conditioning unit (15–110 s)	Visual + histopathological	Optimal decision theory	-	The results demonstrate that, using highly reliable training datasets, the proposed technique is capable of correctly classifying benign and malignant skin lesions with unprecedented accuracy.	100% sensitivity and 90% specificity for 93% of 200 training datasets.
Stringasci et al.[17]	Static IRT	FlukeVR FLK-Ti400 model), image capture frequency of 60 Hz, and 320×240 pixels resolution	-	-	Visual + histopathological	Support Vector Machine	A region including only the lesion itself and a margin of approximately 2 mm from the edge. Statistical attributes (entropy, maximum-minimum subtraction, etc.); histogram analysis; average temperature within and outside the lesion borders, with borders identified by the different filters applied by each attribute ("Skin-lesions interface")	Actinic keratoses and squamous cell carcinoma presented different mean temperatures, while the other pairs of lesions presented similar temperatures. Benign lesions had greater edge detection definition than malignant lesions, as a general rule.	Squamous cell carcinoma x Actinic keratosis was the pair of lesions that had the best discrimination (82.9% sensitivity and 85.7% specificity); Melanoma x pigmented seborrheic keratosis was the pair of lesions with less accuracy (65% sensitivity and 70% specificity)

Table 4 Risk of bias analysis of studies included in present systematic review

Study	Risk of Bias				Applicability concerns		
	Patient selection	Index test	Reference Standard	Flow and timing	Patient selection	Index test	Reference Standard
Godoy et al.[18]	?	?	☺	?	?	☹	☺
Godoy et al.[11]	?	?	☺	?	?	☹	☺
Stringasci et al.[17]	?	?	☺	?	?	☹	☺

Labels: ☺Low risk; ☹High risk; ?Unclear risk

Accordingly, for the specificity outcome, the certainty of the evidence was low, being lowered by one level due to there being a risk of bias and inconsistency and imprecision.

Discussion

Humans are homeotherms; therefore, they maintain their body temperature at a constant level, usually above that of the environment. Heat is transported through the body with the blood acting as the heat transfer medium [19]. Therefore, the skin plays an important role in regulating body temperature [19]. In the presence of pathological alterations, as in the case of tumors, the growth of new capillary blood vessels is induced, producing specific growth factors that promote angiogenesis, which continues through the progression of pre-malignant skin lesions until the development of cancer [11].

Neoplastic tissue is generally associated with increased local blood perfusion and increased metabolic rate, producing a change in skin temperature [20]. Heat transfer by radiation is of great value in medicine because the surface of the human body, the skin, is an efficient radiator that is very close to a perfect black body [21]. Therefore, the emitted radiation is detected by infrared imaging systems that generate temperature readings and a visual representation, through thermograms, of the temperature distribution across a surface [22].

IRT can be categorized into two distinct subtypes: passive IRT and dynamic IRT. Passive IRT involves examining the subject in their natural thermal balance, without the imposition of external thermal stress [23]. In contrast, dynamic IRT allows the analysis of temperature changes over a given time interval. In this imaging method, temperature recording is followed by the application of external stimulus, e.g. cooling or heating stimulus [24]. In this review, only one study used passive IRT [17]. According to Wasilewska [24], the image obtained using passive IRT represents the thermal distribution of a specific moment of image acquisition, which makes this method subject to errors and vulnerable to individual factors, external measurement conditions and technical properties. Furthermore, there is no possibility of evaluating the impact of external stimuli.

Passive IRT only provides 2D thermal images, without timing information. This problem could be solved by periodically acquiring thermal images. However, dynamic IRT always provides temperature measurement information without the need for additional image acquisition. Since temperature differences during thermal recovery to steady state are the focus in dynamic IRT, this method depends, to a lesser extent, on the three main factors noted previously that affect temperature measurement [24].

Regarding thermal stress used in dynamic IRT studies for skin lesions in general, there are a variety of cooling methods, such as immersion in water at 15 °C for 2 min, forced evaporation with alcohol spray and fan air currents [25], cooling by gel, metal medal with 6 s of contact with the skin or compressed air [26] and an aluminum medallion (50 mm diameter and 20 mm height) [27].

Recently, in the study by Verstockt et al. [23], six different cooling techniques were tested on phantom skin surfaces to select the most suitable cooling method for diagnosing skin cancer based on dynamic IRT. In this study, the techniques were divided into three categories based on the heat transfer method: convection cooling, conduction cooling, and evaporative cooling. The authors noted that convection cooling methods (Zimmer Cryo 6 cooler and Ranque-Hilsch vortex tube) offer a more consistent and uniform cooling effect, whereas ice or alcohol for cooling purposes are not suitable when used in conjunction with the IRT. This is mainly because the cooling effects of ice or alcohol can introduce artifacts that affect both the infrared radiation and the view of the thermal camera, thus compromising the accuracy and reliability of the results [23].

Furthermore, Verstockt et al. [23] used a decision matrix to evaluate and compare cooling techniques based on several criteria and after evaluating several factors, the Zimmer Cryo 6 refrigerator was found to be the most suitable cooling method. The Cryo 6 cooler has demonstrated several advantages, including its ability to provide uniform cooling to the skin, consistent cold air conditioning, and high cooling efficiency at -30 °C. Importantly, it did not obstruct the field of view of the thermal camera, allowing accurate recording of the cooling sequence of the skin lesion. Furthermore, the Cryo 6 refrigerator was considered easy to use as it did not require additional equipment or consumables,

reducing the workload during clinical practice, with its disadvantage being the price of the device [23].

The location of the lesion is an important information for differential diagnosis and should be reported in all clinical studies assessing lesion diagnosis. In the study by Baek et al. [28], dynamic IRT, with cold or hot thermal stress, was used to differentiate pigmented basal cell carcinoma from seborrheic keratosis in the head and neck region. In this study, there was a statistically significant difference between the mean velocity constants of pigmented basal cell carcinoma lesions and control sites observed in the cold thermal response graph ($p < 0.001$). However, there were no significant differences in thermal recovery patterns between seborrheic keratosis and control sites for the heat or cold thermal response. The cooling unit used in this study was a thermoelectric refrigerator and a temperature resistance detector. Despite the satisfactory results of this study for skin lesions in the head and neck region, it was not an accuracy study.

In the study by Alves et al. [29], IRT was used comparing with power Doppler for the analysis of nodule vascularization and subsequent selection of nodules to be subjected to fine needle aspiration biopsy. In this study, in relation to the findings of fine needle aspiration biopsy, the sensitivity, specificity, positive predictive value, negative predictive value and accuracy of power Doppler were 95.16%, 23.52%, 96.22%, 16.70% and 89.51%, respectively, compared with 100%, 58.06%, 87.73%, 100% and 89.51%, respectively, for IRT.

To facilitate the understanding of thermal image visualization and optimize the final diagnosis, two studies of the studies assessed in this systematic review explored the application of artificial intelligence classifiers [11, 17]. Artificial intelligence, in various forms and degrees, has been used to develop and promote a wide spectrum of fields, such as banking and financial markets, education, supply chains, manufacturing, retail and e-commerce, and healthcare [30]. As in other imaging modalities, the processing of thermographic images can be performed with artificial intelligence approaches to save time and assist healthcare professionals in different diagnostic task [31]. Artificial intelligence tools have been used in IRT exams in various situations, such as skin lesions detection and differentiation [27, 31], breast cancer detection [32] and diabetic feet assessment [33].

In Magalhaes et al. [27] study, the SVM, ANN and k-NN classifiers were tested to differentiate melanoma lesions (16 samples) and melanocytic nevi (30 samples) based on thermal profiles. In this study, the SVM classifier achieved an accuracy of 84.2% and a sensitivity of 91.3%, but poor specificity (11.1%). In another study by the same group, Magalhaes et al. [31], aiming to explore the potential of infrared thermal imaging as an auxiliary tool in the diagnosis of skin cancer lesions using ensemble learning and deep learning, it was observed that the deep learning strategy outperformed the ensemble learning strategy, showing good predictive performance for differentiating melanoma and nevi (Accuracy = 0.9665, Recall = 0.9411, F1 score = 0.9536, AUC = 0.9185) and melanoma and non-melanoma skin cancer (Accuracy = 0 0.9259, Recall = 0.8852, F1 score = 0.9051, AUC = 0.901).

In their study, Zhang et al. [34] present a novel Adaptive multi-modal Hybrid model to combine the ultrasound and IRT images for the classification of thyroid nodules, que on the collected dataset, AmmH model respectively achieved 97.17% and 97.38% of F1 and F2 scores. The authors further concluded that the model demonstrated promising classification performance, indicating its potential as a non-invasive, radiation-free, and cost-effective screening tool to distinguish between benign and malignant thyroid nodules.

This systematic review evaluated the effectiveness of IRT in differentiating benign and malignant lesions in the head and neck skin region. Although dynamic IRT demonstrated good accuracy for this purpose, the amount of evidence found was reduced. Furthermore, all the studies were considered as at risk of bias with concerns regarding applicability, and the certainty of evidence for the outcomes assessed was very low.

Although promising, the use of artificial intelligence as an aiding tool to assess IRT images still needs to be further investigated and more reliable results should be possible with larger samples. Furthermore, IRT studies involving skin lesions in the head and neck region must be more robust, including all necessary basic methodology information such as patients' sociodemographic data, anatomical location of the lesions, and how thermal stress was applied. This is essential for ensuring the reproducibility of the studies and for the results to be reliable and correctly analyzed.

Strengths and limitations

The strength of this systematic review lies in the fact that it is the first study to investigate the use of IRT in differentiating benign and malignant skin lesions in the head and neck region, with a rigorous process of critical assessment of the evidence using the GRADE approach.

There were limitations in the present systematic review due to several issues found in the identified evidence. Some potential studies could not be included due to lack of any information on the anatomical location of the lesions [27, 31, 35, 36]. However, three studies were included despite lacking information on the anatomical locations of the lesions, as they provided figures of the cases analyzed [11, 17, 18].

Other limitations were identified in the studies, such as reduced or lack of reporting of sample size and problems with the methodological quality.

Implications for clinical practice and research

Based on the results of the evidence found in this review, it seems that dynamic IRT is more effective method in differentiating skin lesions in the head and neck region than the passive technique. However, it is important to note that was not possible to perform subgroup analysis.

Furthermore, the location of the lesion is an important information for differential diagnosis and should be reported in all clinical studies assessing the diagnosis of such lesions. More detailed information such as sample size and sex of individuals should be included in future studies. Furthermore, the use of other thermal methods for IRT, with high methodological rigor, must be used to confirm the evidence.

Conclusion

The studies included in this systematic review have varying results regarding the accuracy of IRT in differentiating lesions with low certainty of evidence. Therefore, more studies with appropriate methodological designs and larger samples are needed to obtain more conclusive results.

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Declarations

Competing interests The authors declare no competing interests.

Informed consent Not applicable.

Ethical approval Not applicable.

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